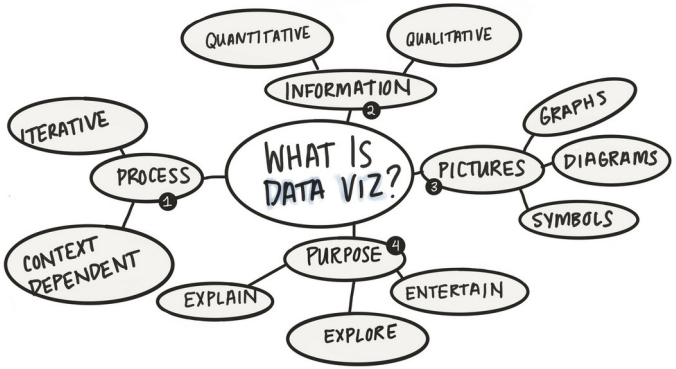
Lecture 18

Visualization Recap



Today

Some takeaways from our look at data viz

Data Analysis Workflow

- Explore data – get a sense of what we do and do not have
- Formulate questions
- Analyze dataset to build evidence through both stats and plots
- Formulate and analyze new hypotheses
- Write up your findings

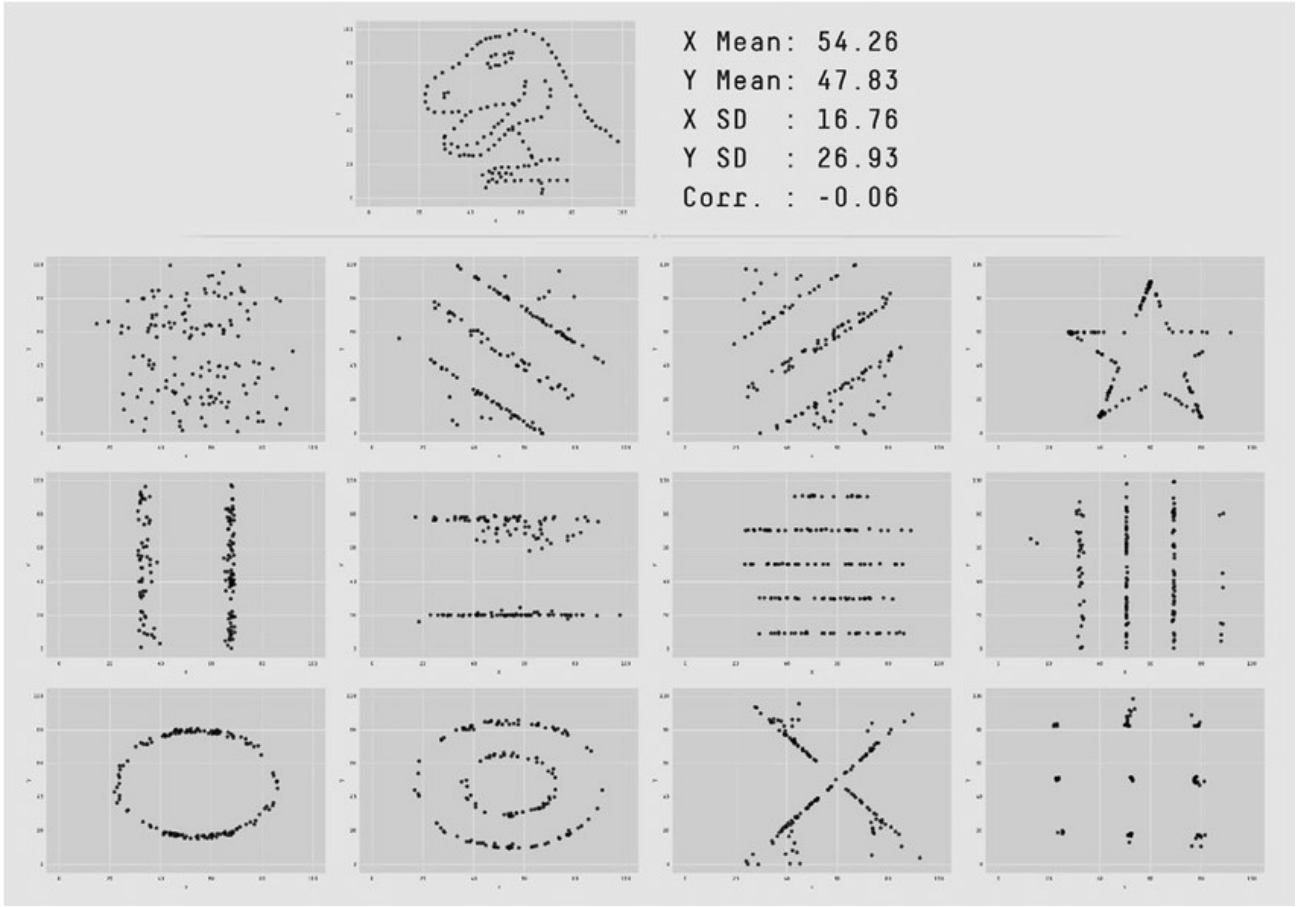
Look at the Data

- We've learned lots of powerful tools to summarize data
  - Counts, means, quantiles, variance, covariance, etc.

- We may think that these are sufficient for data analysis
  - Common shortcut

- Doing so can lead us to miss a lot of structure in the data!

Look at the Data



Respect Human Visual Perception

- **Highest Accuracy**: Position on a common scale (Scatterplots, Bar charts)
- **Medium Accuracy**: Length, Direction, Angle (Pie charts are hard to read because humans struggle to compare angles)
- **Lowest Accuracy**: Area, Volume, Color Saturation (Bubble charts and heatmaps are great for spotting general trends, but terrible for exact value comparisons)

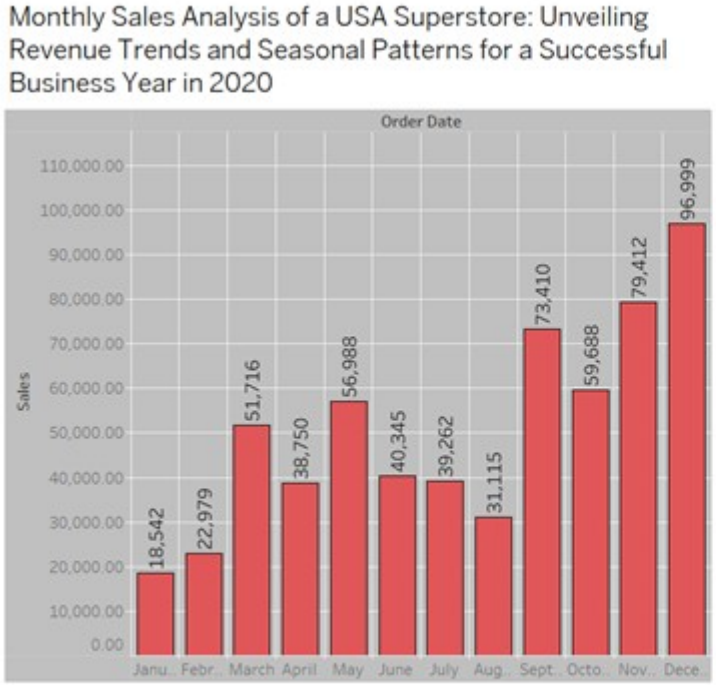
Takeaway: Map your most important variables to the X and Y axes (Position), and reserve color, size, and shape for secondary categorical variables

Maximize the Data-Ink Ratio

Idea: strip away plot elements that do not add new information

- Heavy backgrounds, dark gridlines, 3D effects on 2D data, redundant legends, and unnecessary borders
- These elements increase cognitive load on the reader

Maximize the Data-Ink Ratio



Low Data Ink Ratio



High Data Ink Ratio

Maximize the Data-Ink Ratio

Idea: strip away plot elements that do not add new information

- Heavy backgrounds, dark gridlines, 3D effects on 2D data, redundant legends, and unnecessary borders
- These elements increase cognitive load on the reader

This can go too far!

- Removing too many plot elements can make the plot difficult to interpret

Communicate Context

A plot should make sense even if the audience ignores the surrounding text

- The audience should not have to guess what they are looking at

Takeaways:

- Every plot needs clear axis labels (with units!) and a legend if colors/shapes are used
- Most plots would benefit from a descriptive title

Use Color with Purpose

Color should encode meaning, not just decorate the page

- If changing the color of a bar doesn't change the meaning of the data, the color is usually unhelpful
- Sequential Palettes (e.g., `Blues`): Best for ordered, continuous data (low to high).
- Diverging Palettes (e.g., `coolwarm`): Best for continuous data with a meaningful midpoint, like 0 (negative to positive)
- Qualitative Palettes (e.g., `colorblind`): Best for unordered categorical data (Country A vs. Country B)

Takeaway: Default to a neutral color (like grey) for most data points, and use a bright highlight color intentionally to draw the audience's eye

Honesty and Good Faith

Data visualization is powerful, which means it is easy to lie with it, either accidentally or on purpose.

- Zero Baseline Rule: Bar charts encode value by length. If you truncate the Y-axis so it doesn't start at zero, you visually exaggerate the difference between bars (Scatterplots and line charts do not always need to start at zero, because they encode by position, not length)
- Avoid Dual Y-Axes: Plotting two lines on two different Y-axes with different scales forces an artificial visual intersection that implies correlation where none may exist

Takeaway: Choose the visualization that accurately reflects the magnitude and relationship of the numbers, not the one that looks the most dramatic

Next Time

- Reporting your findings